CANDIDATE'S NAME: ______________________________ **CTG:** _________

YISHUN JUNIOR COLLEGE

2011 JC2 PRELIMINARY EXAMINATION

CHEMISTRY 9647/03
HIGHER 2

Paper 3 Free Response **MONDAY 15 AUGUST 2011**

1400hrs − 1600hrs
(2 hours)

Candidates answer in separate paper.

Additional materials:

Writing Papers and Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and CTG on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together with the cover sheet on top.

This question paper consists of 10 printed pages.

Answer any **four** questions

1 This question is about the reactions involving Period 3 elements.

(a) (i) Describe the reactions, if any, that occur when separate samples of sodium, aluminium and sulfur are burnt in excess air. Write equations where appropriate.

(ii) Describe, with equations for any reactions that occur, the action of water on the products formed in **(a)(i)**. State the effect of adding a few drops of Universal Indicator solution to each resulting solution.

[6]

(b) In the presence of phosphorus, carboxylic acids react with $Cl_2(g)$ to form α–chloroacids, which are good intermediates for the synthesis of amino acids in the laboratory.

$$R\text{–}CH_2CO_2H \xrightarrow{P\ +\ Cl_2} R\text{–}CHClCO_2H$$

(i) Using the reaction given above, suggest how **each** of the following amino acids may be synthesised from suitable carboxylic acids.

- 2-aminopropanoic acid
- 3-aminopropanoic acid

In each case, state the reagents and conditions required and include any intermediate(s) that is formed.

(ii) Phosphorus acts as a catalyst for the reaction. Explain, with reference to the Maxwell Boltzmann energy distribution curve, how the use of catalyst speeds up a chemical reaction.

[8]

(c) Under suitable conditions, SCl_2 reacts with water to produce a yellow precipitate and a solution **J**. Solution **J** is a mixture of $SO_2(aq)$ and compound **K**.

(i) Suggest the identity of compound **K**.

(ii) Write a balanced equation, including state symbols, for the reaction between SCl_2 and water.

(iii) Describe what would be observed when each of the following is added to separate samples of solution **J**. Write balanced equation for any reaction that occurs.

- $AgNO_3(aq)$
- acidified $K_2Cr_2O_7(aq)$

[6]

[Total: 20 marks]

2 **(a)** The following table lists the boiling points of some unbranched alkanes and alcohols.

formula	boiling points / °C
C_2H_6	−88
C_3H_8	−42
C_4H_{10}	0
CH_3OH	65
C_2H_5OH	78
C_3H_7OH	97
C_4H_9OH	118
$C_5H_{11}OH$	133

(i) Explain the trend in the boiling points of the alkanes

(ii) If the intermolecular bonding in ethanol, C_2H_5OH, were to be similar to that of the alkanes, predict a value for its boiling point.

(iii) Explain why the boiling points of the alcohols are much higher than those of the alkanes.

(iv) A mixture containing two of the alcohols listed in the table above, is to be separated using fractional distillation. The mixture is placed in the distillation apparatus at room temperature and then gently heated. The first fraction is collected at 97.2 °C.

I Identify one alcohol from the table above that could not be present in the mixture.

II By specifically referring to this experiment, explain why the alcohol identified in **(I)** could not be present.

III Give a reason why the distillation flask should not be heated using a bunsen burner.

[9]

(b) Nuclear medicine makes wide use of a metastable form of an isotope of technetium, ${}^{99}_{43}Tc$. ${}^{99}_{43}Tc$ is injected into the bloodstream as the compound $NaTcO_4$ to monitor blood flow in the vital internal organs. Technetium and manganese are in the same group of the Periodic Table.

(i) An acidified solution of $NaTcO_4$, may react with iodide to give iodine, and itself reduced to Tc^{2+}. Construct an equation to represent this reaction.

(ii) The standard electrode potential for the reduction of TcO_4^- in acidic medium to Tc^{2+} is +0.85 V.

Outline with the aid of a labelled diagram, how you would determine the standard potential of a TcO_4^-(aq) / Tc^{2+}(aq) electrode system.

[5]

(c) One method that is used to determine the concentration of the ozone layer is to pass air through acidified potassium iodide and to measure the amount of iodine liberated. The following reaction takes place:

$$O_3 + 2H^+ + 2I^- \rightarrow O_2 + H_2O + I_2$$

The iodine liberated is measured using platinum/iodine electrode against standard silver / silver chloride reference electrode. The e.m.f. of the system, in volts, is given by the following equation:

$$E = 0.32 + 0.029\ \lg[I_2]$$
$$(\lg = \log_{10})$$

To determine ozone in the atmosphere above New Zealand, a balloon was used to carry a sampling device. A 1.00 m^3 sample of air, measured at 25 °C and 0.24 atm was passed through 10 cm^3 of acidified potassium iodide. The potential of the platinum electrode dipping into the solution against the standard silver/silver chloride electrode was 0.21 V.

Determine the volume of ozone in this sample of air.

[3]

(d) Cervonic acid is a polyunsaturated monobasic acid commonly found in fatty fish oil. The number of carbon-carbon double bonds in a molecule of cervonic acid can be determined by titration with iodine, I_2, solution.

20.00 cm^3 of 0.300 mol dm^{-3} iodine solution reacted with 0.328 g of cervonic acid. The molar mass of cervonic acid is 328.0 g mol^{-1}.

(i) Calculate the number of carbon-carbon double bonds in a molecule of cervonic acid.

(ii) Given that there are 22 carbon atoms in a molecule of cervonic acid, deduce the molecular formula of cervonic acid.

[3]

[Total: 20 marks]

3 **(a)** The halides react differently with concentrated sulfuric acid.

(i) Describe how the reaction of concentrated sulfuric acid with potassium chloride differs from that of concentrated sulfuric acid with potassium bromide. Suggest an explanation for the differences in reaction and write equations illustrating the types of reaction undergone.

(ii) Astatine is found below iodine in Group VII of the Periodic Table. Write appropriate equation(s) to show the reaction(s) when concentrated sulfuric acid is warmed with solid potassium astatide, KAt.

[5]

(b) 0.010 mol of hydrogen and 0.010 mol of iodine were placed in a 2.0 dm^3 flask. The mixture was heated to 400 °C at 1.1 atm. After an hour, the following equilibrium was established.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

The flask was then rapidly cooled, the HI dissolved in water and made up to 100 cm^3 in a standard flask. Portions of 25.0 cm^3 were extracted and titrated against 0.100 mol dm^{-3} NaOH. The average titration reading was 30.00 cm^3.

(i) Explain why the flask was rapidly cooled.

(ii) Calculate how many moles of HI are present at equilibrium.

(iii) Write the K_c expression for the reaction and calculate its value.

(iv) The experiment was repeated with the same amounts of H_2 and I_2, in the same flask but the temperature was increased to 600 °C. The value of K_c was 0.36.

Determine the composition of the equilibrium mixture.

The enthalpy change ΔH, in J mol^{-1}, for a reaction can be found by measuring K_c at two temperatures, T_1 and T_2. The values are related to each other by the following equation:

$$2.3\lg\left[\frac{K_{c1}}{K_{c2}}\right] = \Delta H\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

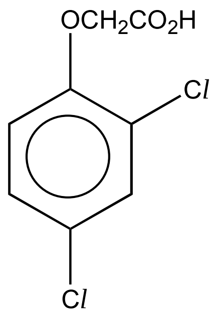
(T_1 and T_2 are measured in Kelvins; lg = $\log_{10}$)

(v) Calculate ΔH by using your value of K_c from **(b)(iii)** and the value of K_c at 600 °C given in **(b)(iv)**.

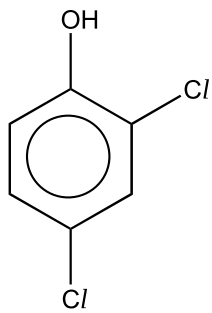
(vi) Given that $\Delta H_f^{\ominus}$ of HI(g) is +27 kJ mol^{-1}, suggest why the ΔH obtained from **(b)(v)** is not twice the value of $\Delta H_f^{\ominus}$ of HI(g).

[9]

(c) 2,4-dichlorophenoxyethanoic acid (sold as 2,4-D) is a powerful herbicide. It mimics a plant growth hormone and causes broadleaved weeds literally to grow themselves to death. American forces used it as a defoliant in Vietnam under the name of *Agent Orange*.



2,4-D



2,4-dichlorophenol

2,4-D is synthesised by treating 2,4-dichlorophenol successively

- with OH^-(aq);
- then with $ClCH_2CO_2Na$;
- finally with H^+(aq).

(i) What is the purpose of OH^-(aq)?

(ii) Outline the mechanism for the stage involving $ClCH_2CO_2Na$, showing clearly the flow of electrons from one species to another.

(iii) Write a balanced equation showing the reaction that took place when H^+(aq) was added.

(iv) Suggest a positive chemical test for **each** of the two compounds dissolved in appropriate solvents that could be used to distinguish 2,4-D and 2,4-dichlorophenol from each other. You should state what you would observe for **each** compound in **each** test.

[6]

[Total: 20 marks]

4 Carbon is the fourth most abundant element by mass in the universe. This element, together with hydrogen and oxygen, forms many acidic compounds.

(a) One example of such an acidic compound is **P**, with the molecular formula $C_8H_{14}O_4$.

When 0.10 mol of **P** reacts with excess anhydrous $SOCl_2$, 0.10 mol of **Q** and 0.20 mol of HCl are formed. **P** gives an orange precipitate on reaction with 2,4-dinitrophenylhydrazine and gives no observable change when heated with acidified $K_2Cr_2O_7$. When **P** is heated under reflux with concentrated sulfuric acid, two different compounds, **R** and **S**, with the same molecular formula, $C_8H_{12}O_3$, are obtained.

On treatment with hot concentrated acidified $KMnO_4$, only **R** can be oxidised to form two acidic compounds **T**, $C_3H_4O_3$, and **U**, $C_5H_8O_3$. **U** is found to be optically inactive.

When heated with alkaline aqueous iodine, all the compounds, **P**, **Q**, **R**, **S**, **T** and **U** produce a yellow precipitate with an antiseptic smell.

Using the information given above, identify the organic compounds **P**, **Q**, **R**, **S**, **T** and **U**. Explain your reasoning.

[10]

(b) Atmospheric carbon dioxide dissolves in water to form a weakly acidic solution containing the hydrogencarbonate ions, HCO_3^-.

$$CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+ \qquad K_a = 7.94 \times 10^{-7} \text{ mol dm}^{-3}$$

(i) Draw a dot-and-cross diagram to illustrate the bonding in the hydrogencarbonate ion. Your diagram should include the outer electrons only.

(ii) State the hybridisation of the carbon atom in the hydrogencarbonate ion.

The CO_2/HCO_3^- system is the main buffering agent in blood plasma.

(iii) Explain how this system acts as a buffer, illustrating your answer with relevant equations.

(iv) Determine the pH of normal blood plasma if $[CO_2] : [HCO_3^-] = 1 : 20$.

[5]

(c) The element chlorine can also be found in some carbon-containing acids.

The table below shows the pK_a values of three carboxylic acids, two of which contain chlorine.

acid	formula of acids	pK_a
I	$CH_3CH_2CO_2H$	4.9
II	$CH_2ClCH_2CO_2H$	4.1
III	$CH_3CHClCO_2H$	2.8

(i) Calculate the pH of 0.10 mol dm^{-3} solution of acid III.

(ii) Explain the trend of decreasing pK_a values of the three acids.

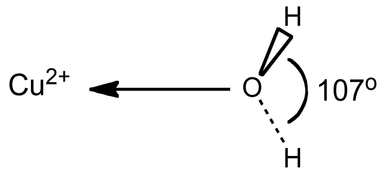
[5]

[Total: 20 marks]

5 **(a)** Copper(II) sulfate pentahydrate, $CuSO_4.5H_2O$, is a fungicide. When mixed with calcium oxide it is called the Bordeaux mixture and is used to prevent fungal diseases in grapes, melons and other berries.

(i) Write the electronic configuration for the copper(II) ion and use it to explain why copper is a transition element.

(ii) Aqueous copper(II) sulfate contains $[Cu(H_2O)_6]^{2+}$ ions. The diagram shows part of the $[Cu(H_2O)_6]^{2+}$ ion and the H-O-H bond angle in the water ligand.



Explain why the H-O-H bond angle in the water ligand is 107° rather than 104.5°.

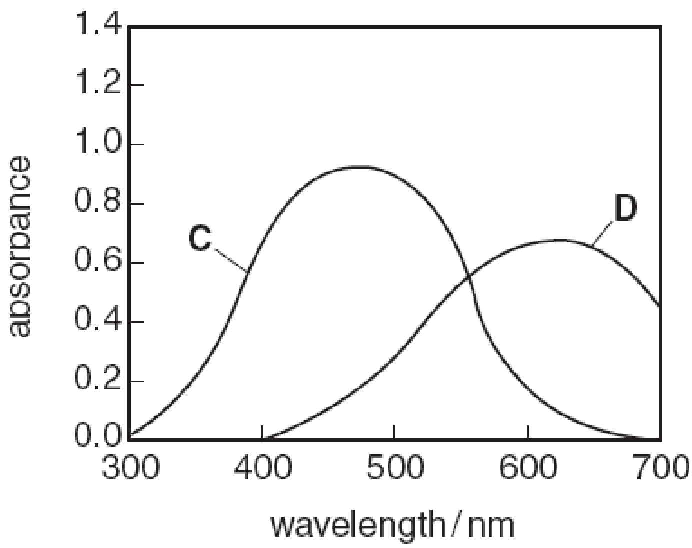
[4]

(b) Aqueous solutions containing complexes of transition metals are usually coloured. This is due to the absorption of part of the spectrum of white light passing through the solution.

The following table lists the colours and energies of photons of light of certain wavelengths.

wavelength / nm	energy of photon	colour of photon
400	high	violet
450	↓	blue
500	lower	green
600	↓	yellow
650	low	red

The visible absorption spectra of solutions of two transition metal complexes **C** and **D** are shown in the diagram below.



(i) What are the colours of the solutions of the two transition metal complexes **C** and **D**?

(ii) In which complex, **C** or **D**, is the energy gap between the two groups of 3d orbitals larger? Explain your answer.

[3]

(c) *Aqua regia* is a mixture of one part of concentrated nitric acid and three parts of concentrated hydrochloric acid, This mixture is used to dissolve inert metals such as gold and copper. The overall reaction between *aqua regia* and gold is as follows:

$$4H^+(aq) + 4Cl^-(aq) + Au(s) + NO_3^-(aq) \rightarrow AuCl_4^-(aq) + NO(g) + 2H_2O(l)$$

Given that the reaction takes place in two steps, explain how gold is dissolved in *aqua regia*.
[3]

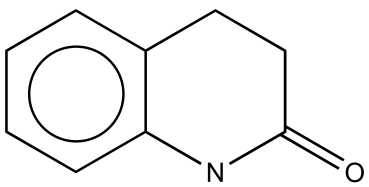
(d) A compound **E** has the molecular formula, $C_3H_4OCl_2$. Compound **E** is a colourless liquid which fumes in moist air. It reacts with cold water to form compound **F**, $C_3H_5O_2Cl$, which is optically active. When compound **E** is heated under reflux with aqueous sodium hydroxide and then acidified, compound **G**, $C_3H_6O_3$, is obtained.

(i) Deduce the structural formulae of compounds **E**, **F** and **G** and explain the reactions that occurred.

(ii) Starting from ethanol, devise a synthesis pathway to prepare compound **G**. You should state the reagents and the reaction conditions that are needed for each stage.

[8]

(e) Suggest a chemical test, stating the reagents and observations that enable the compounds below to be distinguished from each other.



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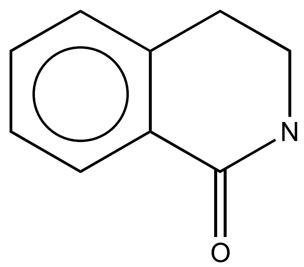

[2]

[Total: 20 marks]

~ END OF PAPER ~